Workplace Demands and English-Language Participation: An Exploratory Longitudinal Trace Analysis of Occupational English Learners

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ABSTRACT

This paper presents an exploratory longitudinal trace analysis of KakaoTalk instructional message records from four adult professional English learners across 269 to 824 days of instruction. Drawing on behavioral trace analytics and self-regulated learning theory, we examine whether scheduling-disruption language in learner messages co-varies with apology and uncertainty markers — a proxy for communicative stance under occupational constraint. Within-learner standardised mean differences (SMDs) show that disruption-coded weeks are associated with elevated apology/uncertainty language for three of four learners (SMD range: +0.46 to +0.78), with the strongest learner showing stability across keyword sensitivity checks (SMD: +0.75 to +0.81). English-message proportion time trends diverge across learners, with one showing a positive slope with confidence interval entirely above zero. These results are exploratory, grounded in a small-N corpus, and interpreted as patterns warranting more rigorous study rather than as population-level findings. The pipeline — manifest-driven parsing, learner-week aggregation, within-learner sensitivity analysis — is fully reproducible and is offered as a methodological model for practitioner-originated trace corpora.

Keywords: learning analytics, occupational English learning, self-regulated learning, behavioral traces, longitudinal corpus, KakaoTalk

1. INTRODUCTION

Adult professionals studying a second language face a participation structure that is qualitatively different from that of classroom learners. They attend sessions irregularly, reschedule under occupational pressure, and produce language practice in compressed windows between meetings and deadlines. For these learners, the question of when and whether to engage with instructional material is itself a self-regulation problem — a negotiation between learning goals and workplace demands that plays out in real time across weeks and months.

The instructional messaging record — exchanges between learner and instructor over digital communication platforms — is a natural site where this negotiation is visible. Rescheduling requests, scheduling-friction language, apologetic messages, and correction exchanges leave systematic traces in a chronologically ordered text corpus that persists far beyond the individual session. Yet practitioner-collected instructional records of this kind have been largely absent from the Learning Analytics literature, which has tended to study institutional learning management systems, MOOCs, or controlled experimental platforms.

This paper reports an exploratory analysis of KakaoTalk message records from four adult professional English learners enrolled in private instruction across technology, automotive, and consumer goods firms in South Korea. The corpus spans 6,045 parsed messages and 327 observed learner-weeks. We ask: (1) does scheduling-disruption language in learner messages co-vary with apology and uncertainty markers within individual learners? (2) How do English-message participation trends differ across learners over the observation window? And (3) are these patterns stable when the dominant keyword terms are removed?

We make two contributions. First, empirically: we provide within-learner longitudinal evidence consistent with SRL theory's prediction that contextual demands co-vary with learner communicative stance, with the strongest association stable across keyword sensitivity tests. Second, methodologically: we describe a manifest-driven, reproducible parsing and analysis pipeline for practitioner-collected chat corpora that makes the analytical choices transparent and auditable.

2. BACKGROUND

2.1 Behavioral Trace Analytics and the LA Feedback Loop

Clow (2012) describes the learning analytics cycle as a loop connecting data, metrics, interventions, and learner response. The motivating claim is that digital traces left by learning activity — click streams, forum posts, submission patterns — can surface patterns that are invisible to instructors managing many learners simultaneously. Most trace analytics research has focused on formal institutional data systems. Practitioner-collected records from informal instructional settings represent an underexplored variant of the same data type: longitudinal, ecologically valid, and rich with real communicative context, but lacking the standardisation of LMS log data.

2.2 Self-Regulated Learning Under Occupational Constraint

Zimmerman (2000) defines self-regulated learning as metacognitive, motivational, and behavioral activity in the service of learning goals. Pintrich (2000) identifies four regulation domains: cognition, motivation, behavior, and context. Context regulation — managing environmental demands to create conditions for learning — is particularly relevant for adult occupational learners, who must schedule study time around shifting professional obligations. Irregular participation, scheduling requests, and apologetic framing may be behavioral and communicative traces of context regulation under pressure.

2.3 Communicative Anxiety and Apology in L2 Settings

Horwitz, Horwitz, and Cope (1986) established foreign language anxiety as a construct associated with communication apprehension and fear of negative evaluation. In written instructional messaging, hedging and apologetic language may reflect related communicative stances — though Brown and Levinson (1987) caution that apology and deference are routine politeness strategies in Korean professional discourse, independent of affective state. This cross-cultural ambiguity means that Korean learner apology markers must be interpreted with particular care, and motivates the sensitivity analysis reported in Section 4.3.

2.4 Corrective Feedback in Instructional Interaction

Lyster and Ranta (1997) demonstrated that corrective feedback type differentially affects learner uptake. Heift (2010) showed that even in CALL contexts, the presence of corrective prompts influences engagement. In this corpus, instructor corrections appear as explicit keyword-coded notation (correction, suggestion, ->, =>) and as reformulation-style corrections prefixed with >. These two formats are not equally distributed across learners — a formatting habit of the instructor rather than a learner difference — and are tracked separately to preserve cross-learner comparability.

3. METHODS

3.1 Corpus

The corpus consists of KakaoTalk export files from four private English instruction relationships, collected between February 2024 and May 2026. All four learners are adult professionals at Korean firms in technology, automotive, and consumer goods sectors. The instructor is the same person across all four relationships. Raw exports are private; the analysis pipeline is publicly available and reproducible from a synthetic example export.

Table 1. Corpus Summary

Learner	Occupational domain	Obs. weeks	Learner msgs	Instr. msgs	Span (days)
L001	Automotive / manufacturing	31	74	146	268
L002	Technology / engineering	104	354	460	815
L003	Consumer goods / corporate	79	1,179	1,618	575
L004	Technology / software	113	595	1,619	824

Learner identifiers are anonymised. Observation windows are not designed to be comparable.

3.2 Parsing and Aggregation

A manifest-driven R parser reads each KakaoTalk export, identifies messages by matching the [sender][time] header pattern, and aggregates to a learner-week summary table. The parser handles continuation-line merging (multi-line messages appear as single header rows in the export), tracks > prefix corrections separately from keyword-coded corrections, and logs raw line counts, date spans, and unmatched lines per export for transparency. Post-audit revisions removed the standalone English term work from the disruption keyword list after inspection revealed systematic false positives on lesson content. All parsing and analysis scripts, along with a synthetic example export containing no real learner data, are publicly available.

3.3 Indicators

Three indicator variables are used in the analysis. The schedule-disruption marker flags learner messages containing scheduling-friction language (English: meeting, busy, late, overtime, business trip, deadline, delay, reschedul, can't make, work trip; Korean: 즛회의, 야근, 출장, 일정, 변경, 불가, 늘). The apology-uncertainty marker flags learner messages containing apology, hedging, or uncertainty language (English: sorry, maybe, not sure, i think, i guess, afraid, worried, can't, cannot; Korean: 죄송, 미안, 걱정, 모르, 어렵, 못). The correction-density variable counts instructor messages containing explicit correction notation per instructor message. All three are text-based proxies, not validated instruments. Their theoretical rationale and known false positive risks are documented in a companion indicator validation memo.

3.4 Analysis Strategy

All analyses are within-learner or learner-fixed-effect screens. Pooled estimates are never reported without first presenting individual learner slopes or effect sizes. Within-learner standardised mean differences (SMDs) compare apology-uncertainty rates between disruption-coded and non-disruption-coded weeks. Within-learner OLS slopes estimate time trends per learner. A sign-agreement check across learners precedes any pooled summary. Sensitivity checks rerun the primary SMD analysis under four keyword definitions: broad (all keywords), no-sorry (removing the dominant English apology term), no-late (removing the dominant English disruption term), and Korean-only disruption.

4. RESULTS

4.1 Participation Heterogeneity Across Learners

Figure 1 shows learner-week message volume and English-message proportion for each learner over their observed period. Observation windows differ substantially (31 to 113 weeks), which limits direct cross-learner comparison. Within learners, participation is irregular: all four show weeks with no observed messages, with gaps attributable to scheduling breaks, travel, or weeks without instructional messaging. Mean weekly learner message counts range from 2.4 (L001) to 14.9 (L003), and English-message proportion ranges from 0.56 (L002) to 0.94 (L004).

Disruption-coded weeks (marked in Figure 1) are not uniformly distributed across the observation period for any learner. L003 and L004 show the highest disruption week rates (48% and 39% respectively), though sensitivity analysis reveals this is substantially driven by the term late. Removing late reduces L003 disruption weeks from 38 to 20 and L004 from 44 to 14. Korean-only disruption keywords produce only 2 disruption weeks for L003 and zero for L004, indicating that these learners communicated scheduling friction primarily in English rather than in Korean scheduling vocabulary.

4.2 Disruption-Coded Weeks and Apology/Uncertainty Language

Table 2 reports within-learner SMDs comparing apology-uncertainty marker rates between disruption-coded and non-disruption-coded weeks. Three of four learners show positive SMDs, consistent with the prediction that scheduling-friction weeks co-occur with elevated hedging and apologetic language. L001 shows a negative SMD, but has only four disruption-coded weeks; this estimate should be treated with substantial caution.

Table 2. Within-Learner SMDs: Apology/Uncertainty Rate, Disruption vs. Non-Disruption Weeks

Learner	SMD (broad)	SMD (no-sorry)	SMD (no-late)	SMD (Korean-only dis.)	n dis / non
L001	-0.56	-0.56	-0.55	-0.54	4 / 27
L002	+0.78	+0.75	+0.81	+0.81	17 / 87
L003	+0.46	+0.49	+0.31	+0.38	38 / 41
L004	+0.55	+0.43	+0.42	NA	44 / 69

NA = no disruption weeks under Korean-only definition (L004). L001 estimate unreliable (n=4 disruption weeks).

L002 shows the most robust finding: the positive SMD (+0.78) is stable across all four keyword definitions, ranging from +0.75 to +0.81. This stability is the strongest available evidence that L002's disruption-apology association is not an artifact of a single ambiguous keyword. The pooled learner fixed-effect model (apology-uncertainty rate ~ disruption-flag + learner-id) yields an estimate of +0.086 (95% CI: [0.041, 0.132]), but this summary conceals L001's negative direction and is reported for context only.

L004's elevated apology-uncertainty rate (mean 0.159 vs. 0.055-0.104 for other learners) deserves particular attention. Removing sorry reduces L004's mean weekly rate from 0.159 to 0.063 — a 60% reduction. L004 uses sorry frequently in contexts of sympathy ("Sorry to hear") and enthusiasm ("Can't wait"), consistent with Korean professional messaging norms where sorry functions as a routine politeness marker. This finding illustrates the cross-cultural interpretive challenge that keyword-based apology markers face in Korean professional contexts.

4.3 English-Message Participation Trends

Within-learner OLS slopes on English-message proportion over study week are shown in Figure 2 (forest plot). Slopes disagree in sign across learners: L003 shows a positive trend (+0.0026 per week, 95% CI: [+0.0010, +0.0042]), the only learner with a confidence interval entirely above zero. L001 and L002 show small negative trends; L004 is near zero. The pooled slope (+0.0001) crosses zero and is not interpretable as a general pattern.

L003's positive trend is the most analytically notable longitudinal finding in the corpus: English-language participation in instructional messaging increases consistently over 79 observed weeks. Whether this reflects increasing English proficiency, increasing comfort with English-medium instructor interaction, or a shift in instructional format cannot be determined from the messaging record alone. These interpretations are distinguishable with supplementary data (OPIc assessments, instructor session logs) but not from text traces alone.

5. DISCUSSION

5.1 What the Findings Mean

The disruption-apology association for L002, the most observation-rich learner, is consistent with SRL context-regulation theory: when scheduling demands intrude, learners' communicative stance shifts toward more hedging and apologetic language. This is the pattern one would expect if scheduling friction creates friction in the learner-instructor relationship — the learner must manage not only the practical constraint but also the face-threatening act of missing or abbreviating a session. The stability of L002's SMD across keyword definitions makes this the strongest finding in the corpus.

The divergence in English-message trends across learners challenges any pooled claim about English-use development. That L003 shows a consistent upward trend while L001 and L002 trend slightly downward suggests that participation trajectories are individual phenomena, shaped by factors that are not visible in the trace record. This finding has a direct implication for LA systems design: aggregate dashboards showing average English-use trends over time would mask trajectories that vary in direction across learners.

5.2 Methodological Contributions

The methodological contribution of this study is the parsing and analysis pipeline for practitioner-collected chat corpora. Three design principles are demonstrated. First, within-learner analysis precedes pooled analysis: SMDs and slopes are computed per learner before any pooled estimate is presented, and sign disagreement is flagged explicitly. Second, sensitivity analysis is built into the reporting pipeline, not added as an afterthought: the R script tests four keyword definitions and reports which findings are robust and which are fragile. Third, the parsing log — recording raw line counts, continuation merges, unmatched lines, and date spans per export — makes the data ingestion layer auditable rather than opaque.

5.3 Limitations

Several limitations constrain the conclusions that can be drawn. The corpus is small (N=4 learners), observation windows are uneven and not designed to be comparable, and the indicators are text-based proxies without inter-rater reliability checks or criterion validity. The disruption-apology association is correlational and cannot be given a causal interpretation. L003 and L004's disruption signals are largely driven by the English term late, which may reflect messaging style or proficiency rather than occupational disruption per se. A full validation study would require experience-sampling for disruption, a validated foreign language anxiety scale for the apology/uncertainty marker, and session records establishing what proportion of instructional feedback appears in written messages.

6. CONCLUSION

This paper demonstrates that practitioner-collected instructional messaging records, handled with appropriate analytical discipline, can generate credible exploratory findings about longitudinal participation patterns in adult occupational language learning. The disruption-apology association for L002 is the most robust finding: disruption-coded weeks show consistently higher apology/uncertainty language regardless of which keywords are included or excluded. English-message participation trends are heterogeneous across learners, with L003 showing a consistent upward trajectory over 79 weeks. These findings are offered as patterns worth investigating in a more rigorously designed study — one with validated instruments, larger N, and a pre-registered disruption operationalisation — rather than as established claims about adult occupational learning. The analysis pipeline is offered as a reproducible model for other practitioner-researchers working with similar data.

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